

Configure IPv6 Addressing

Task – 1 : Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Solution : Assign and Verify a Static IPv6 Address

Step 1: Find the Network Adapter

Open **Command Prompt as Administrator** and run:

```
netsh interface ipv6 show interfaces
```

Step 2: Assign a Static IPv6 Address

Assign : 2001:db8:100:1::10/64

Command : netsh interface ipv6 add address interface="Wi-Fi"
address=2001:db8:100:1::10/64

Step 3: Verify the Address

Run : ipconfig

Look under the Wi-Fi adapter for : IPv6 Address : 2001:db8:100:1::10

You can also use : ipconfig /all

Task – 2 : Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Hint : Use the ping6 or ping -6 command for IPv6 testing.

Solution : Configure a Unique Local IPv6 Address

IPv6 addresses use : fc00::/7

In normal private-network use, they commonly begin with : fd

Step 1: Choose an IPv6 Subnet

For example : fd12:3456:789a:1::/64

Assign our computer : fd12:3456:789a:1::10/64

Command : netsh interface ipv6 add address interface="Wi-Fi"
address=fd12:3456:789a:1::10/64

Step 2: Configure Another LAN Device

Suppose another computer on the same LAN has : fd12:3456:789a:1::20/64

Both devices are in : fd12:3456:789a:1::/64

Step 3: Test Connectivity

To other computer : ping -6 fd12:3456:789a:1::20

Task – 3 : Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Constraint : Document the steps and the observed address behavior in a text file.

Solution : **Temporary IPv6 Address**

There are two concepts worth separating here:

- **Temporary IPv6 addresses** used for privacy are normally automatically generated by Windows when IPv6 privacy addressing is enabled.
- A manually added address can also be made **temporary for the active configuration** using Windows networking commands.

Step 1: View Current IPv6 Addresses

Run : ipconfig

Step 2: Enable IPv6 Privacy Addresses

Run Command Prompt as Administrator : netsh interface ipv6 set privacy state=enabled

disconnect and reconnect Wi-Fi, or restart the computer.

Check again : ipconfig

Step 3 : Compare the Addresses

If Windows generates a new temporary address after the previous address expires or the networking state changes.

Example Observation

Before reconnect:

Temporary IPv6 Address:

2001:xxxx:xxxx:xxxx:aaaa:bbbb:cccc:1111

After reconnect:

Temporary IPv6 Address:

2001:xxxx:xxxx:xxxx:dddd:eeee:ffff:2222

Task – 4 : Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

Solution : **Zomato-Style IPv6 Restaurant Search**

Python Program

```
restaurants = {  
    "Zomato Cafe": "fd12:3456:789a:1::101",  
    "Spice Garden": "fd12:3456:789a:1::102",  
    "Pizza Corner": "fd12:3456:789a:1::103",  
    "Burger House": "fd12:3456:789a:1::104",  
    "Green Bites": "fd12:3456:789a:1::105"  
}  
  
search = input("Enter restaurant name: ").lower()  
  
found = False  
  
for name, ipv6 in restaurants.items():  
    if search in name.lower():  
        print(f"{name} -> {ipv6}")  
        found = True  
  
if not found:  
    print("Restaurant not found.")
```

Example Execution

Enter restaurant name: pizza

Pizza Corner -> fd12:3456:789a:1::103

How It Works

User enters search



Python checks restaurant names



Partial name matches



Restaurant name + IPv6 address displayed

Task – 5 : Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

Solution : **IPv6 Subnetting Scenario**

AI-Generated IPv6 Scenario

Suppose a food delivery company has been assigned : 2001:db8:1234::/48

The company wants three separate networks:

1. **Delivery Department**
2. **Restaurant Department**
3. **Administration Department**

IPv6 LANs commonly use a /64 subnet.

Therefore, we can create:

Department	IPv6 Subnet
Delivery	2001:db8:1234:1::/64
Restaurants	2001:db8:1234:2::/64
Administration	2001:db8:1234:3::/64

Manually Assign an Address

Suppose we choose the **Administration** subnet : 2001:db8:1234:3::/64

We can assign the computer : 2001:db8:1234:3::10/64

Windows Command

```
netsh interface ipv6 add address interface="Wi-Fi"  
address=2001:db8:1234:3::10/64
```

Verify : ipconfig

Expected : IPv6 Address:2001:db8:1234:3::10